AMAZON BOOK RATING NETWORK MODELING & ANALYSIS

MODELING & AI

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INTRODUCTION

- Problem:
  We explore the task of link prediction in a user–book bipartite network derived from Amazon book reviews.
- Objective:
  Compare the performance of traditional heuristic link prediction algorithms vs. a Graph Convolutional Network (GCN).
- Dataset:
  Amazon Book Reviews dataset (Kaggle); ~1.2M nodes, ~2.1M edges.

MOTIVATION: *WHY STUDY LINK PREDICTION ON AN AMAZON-STYLE BOOK NETWORK?*

- Personalized discovery:
  Accurate link prediction → better book recommendations, higher user engagement, and measurable revenue lift [1].
- Cold-start & long-tail challenge:
  In the Amazon Books dataset, > 85 % of users have rated fewer than three books and > 70 % of books have < 10 ratings [2]—an extreme sparsity that hinders traditional recommenders.
- Bipartite complexity:
  Common-neighbor heuristics assume overlapping neighborhoods; this assumption fails in bipartite graphs, motivating path-based scores (Katz) or learned message-passing models (GCNs) [3].
- Explainable vs. learned scores:
  Heuristics give transparent signals; GCNs can capture higher-order structure without manual features—comparing them reveals accuracy/interpretability trade-offs [4].

MODEL(S) USED: HEURISTIC BASED VIA NETWORKX

- Jaccard Coefficient – based on shared neighbors
- Adamic–Adar Index – penalizes common high-degree neighbors
- Resource Allocation Index – similar to Adamic–Adar with stronger penalty
- Preferential Attachment – scores by multiplying node degrees
- Katz Centrality – scores paths of all lengths with exponential decay
- Easy to compute
- Assumes meaningful shared neighbors

MODEL(S) USED: GRAPH CONVOLUTIONAL NETWORK (GCN)

- Implemented in PyTorch Geometric
- Nodes: user and book IDs (no features)
- Layers: 64-dimensional node embedding layer, GCNConv(64 → 128) → ReLU → Dropout, GCNConv(128 → 128), Link decoder: dot product between node embeddings
- Trained with:
  - Binary Cross Entropy loss
  - Adam optimizer, 100 epochs, fixed negative sampling
- Learns from graph structure
- No node features → limits performance

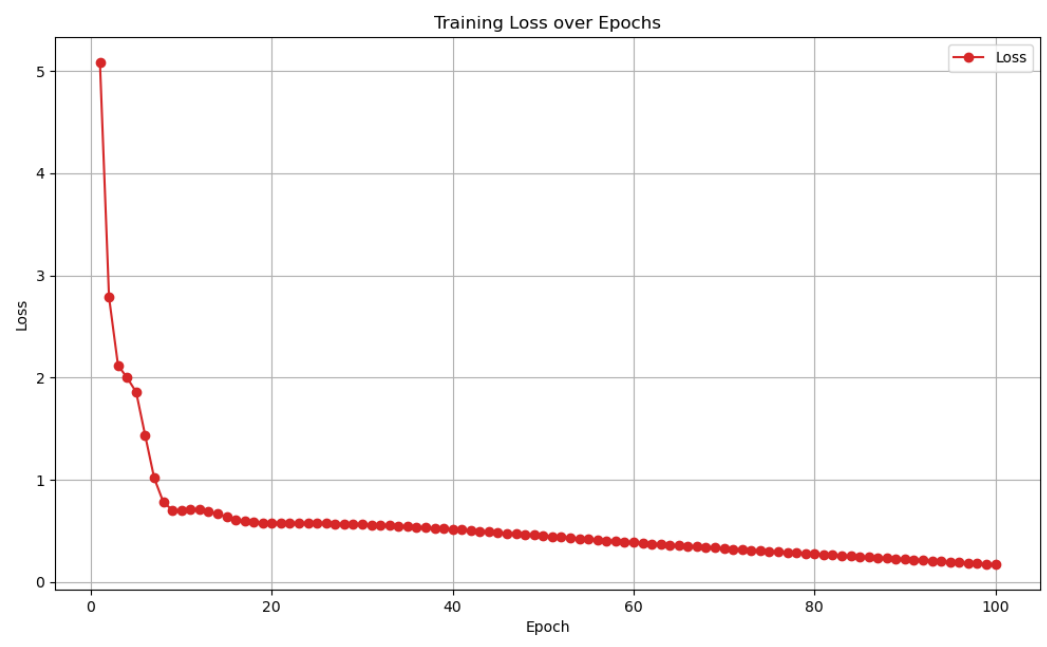
METHODOLOGY

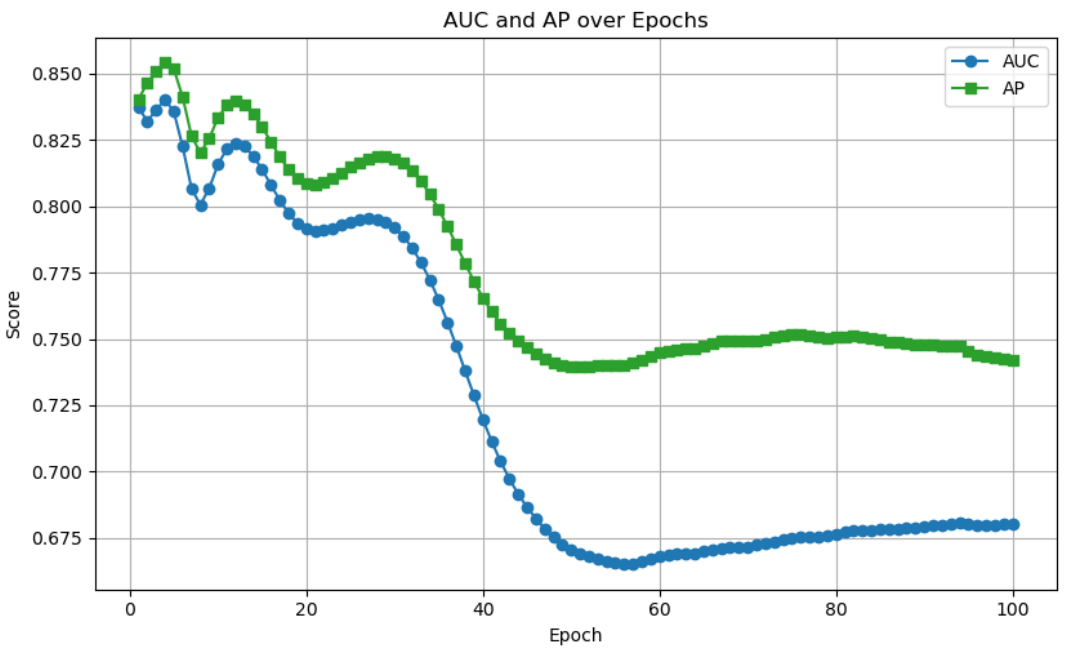
- 1. Build Graph
  User–book bipartite graph from review data
- 2. Split Data
  80/20 train/test edge split + sampled negatives
- 3. Heuristics
  Compute scores with Jaccard, Adamic–Adar, RA, Preferential Attachment, Katz
- 4. GCN Model
  Train 2-layer GCN with embeddings, dropout, and BCE loss
- 5. Evaluation
  Compare AUROC & AP across models

KEY RESULTS

- We evaluated both heuristic-based and learning-based link prediction methods using AUROC and Average Precision (AP).
- Katz Centrality significantly outperformed other heuristics and even beat our GCN model, despite its simplicity.
- Common neighbor–based heuristics like Jaccard and Adamic-Adar failed on this bipartite graph, scoring near random.
- Our 2-layer GCN showed early promise, peaking around epochs 5–10, but suffered from overfitting due to:
  - Lack of node features
  - High graph sparsity
  - No early stopping or regularization beyond dropout
- Katz vs. GCN: In this setup, Katz was more robust—a reminder that classical methods can still outperform deep models in certain conditions.

KEY RESULTS





Method	AUROC	AP
Katz Centrality	**0.8634**	**0.8921**
Preferential Attachment	0.6660	0.7949
Jaccard Coefficient	0.5000	0.5000
Adamic-Adar Index	0.5000	0.5000
Resource Allocation Index	0.5000	0.5000

CONCLUSION

- Katz Centrality performed best overall (AUROC 0.86, AP 0.89)
- GCN model outperformed random guessing but plateaued (AUROC ~0.675)
- Shared-neighbor heuristics failed in sparse bipartite setting
- GCN showed potential despite lack of node features
- Future work: Add user/book features, tune GCN, try GraphSAGE or GAT

REFERENCES

- [1] Gomez-Uribe & Hunt, “The Netflix Recommender System,” *ACM TKDD*, 2016.
  [2] Dataset statistics computed from the Amazon Books Reviews dataset (Kaggle, 2024).
  [3] Kunegis et al., “The Link Prediction Problem in Bipartite Networks,” arXiv:1006.5367, 2010.
  [4] Kipf & Welling, “Semi-Supervised Classification with Graph Convolutional Networks,” ICLR 2017.